Arbitrarily Long Runs in OEIS A202137

An explicit Chinese remainder theorem proof

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Abstract

The sequence A202137 consists of positive integers k for which 24k+1 is neither prime nor a perfect square. We prove that it contains infinitely many runs of every prescribed finite length. The proof forces a selected prime to divide each relevant number to precisely the first power. This gives a self-contained derivation of an assertion already identified as provable in the OEIS.

1. Statement

Theorem. For every integer m >= 1, infinitely many positive integers K satisfy: for each j = 0,1,...,m-1, the integer 24(K+j)+1 is composite and is not a perfect square. In particular, K,K+1,...,K+m-1 are consecutive members of A202137.

2. Proof

Fix m >= 1. Choose distinct primes q_0,...,q_(m-1), each greater than 3. Because 24 is invertible modulo q_j^2, the linear congruence 24(K+j)+1 = q_j (mod q_j^2) has a unique solution for K modulo q_j^2, for each j.

The moduli q_j^2 are pairwise coprime. The Chinese remainder theorem supplies an integer R that satisfies all these congruences simultaneously. Every integer K = R+tM, with M the product of the q_j^2 and t any integer, is also a solution.

For each j, let N_j = 24(K+j)+1. Since N_j = q_j (mod q_j^2), the prime q_j divides N_j, but q_j^2 does not. Thus q_j occurs in the prime factorization of N_j to exponent exactly one; N_j is not a square.

For all sufficiently large positive t we also have K >= 1 and N_j > q_j for every j. Hence q_j is a proper prime divisor of N_j, so N_j is composite. Therefore K through K+m-1 belong to A202137. As t grows these yield infinitely many distinct runs. QED.

3. Scope and verification

OEIS A202137 already records a 2018 note that the run claim follows from the Chinese remainder theorem and growing gaps between squares. This note gives a direct alternative CRT proof; it makes no claim of mathematical novelty or publication priority. The congruence deductions were checked during preparation but not independently peer reviewed.

4. AI authorship and references

Primary author (AI): ChatGPT (OpenAI). Mathematical drafting and preliminary proof checking were performed using AI; no external independent proof certification is claimed.

[1] OEIS Foundation, A202137, https://oeis.org/A202137 (consulted 11 November 2026).

[2] Chinese remainder theorem (classical theorem on coprime moduli).